University of Liberal Arts Bangladesh

Department of Computer Science & Engineering

STA 2101: Statistics & Probability

Lecture 9:
Introduction to Probability

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Class Code

Sec 1 - es2uztok

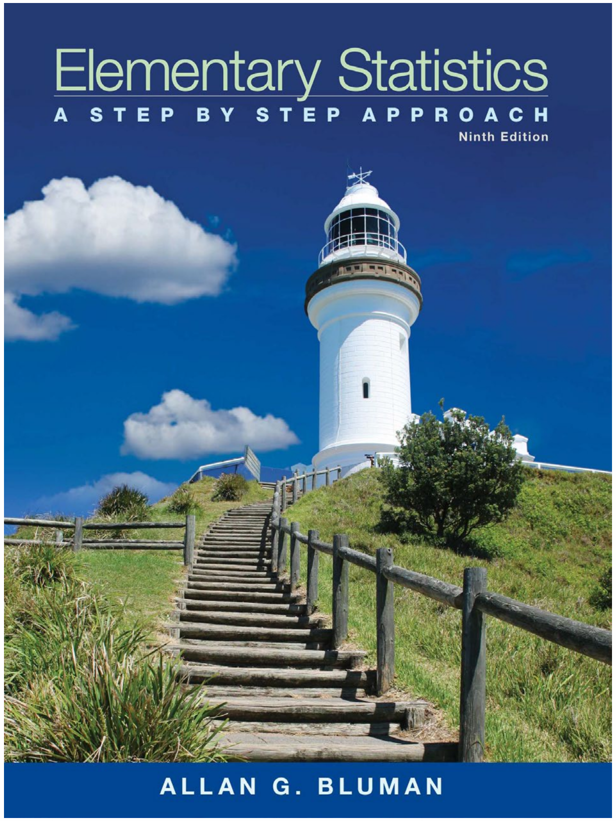
Sec 5 - ayvzsvgm

Spring 2026

Reference Book

- **Text Book(s)**:
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver

- **Reference Material/Book(s)**:
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

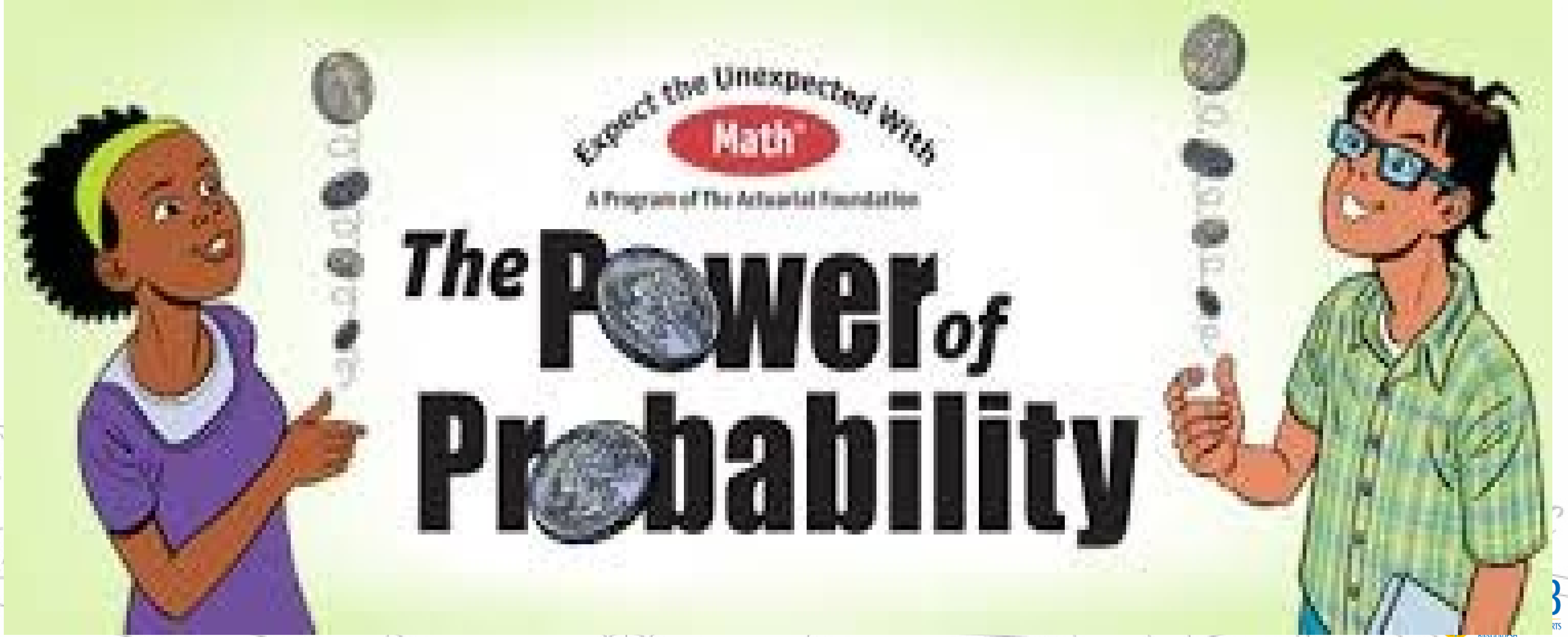
Today's Topics

- Probability

Probability

How likely something is to happen.

Many events can't be predicted with total certainty. The best we can say is how likely they are to happen, using the idea of probability.



Tossing a Coin

When a coin is tossed, there are two possible outcomes:

heads (H) or
tails (T)

We say that the probability of the coin landing H is ½.

And the probability of the coin landing T is ½.

Throwing Dice

When a single die is thrown, there are six possible outcomes: 1, 2, 3, 4, 5, 6.

The probability of any one of them is 1/6.

Probability

In general:

Probability of an event happening = $\dfrac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$

Example: **the chances of rolling a "4" with a die**

Number of ways it can happen: 1 (there is only 1 face with a "4" on it)

Total number of outcomes: 6 (there are 6 faces altogether)

So the probability = 1/6

Example: **there are 5 marbles in a bag: 4 are blue, and 1 is red. What is the probability that a blue marble will be picked?**

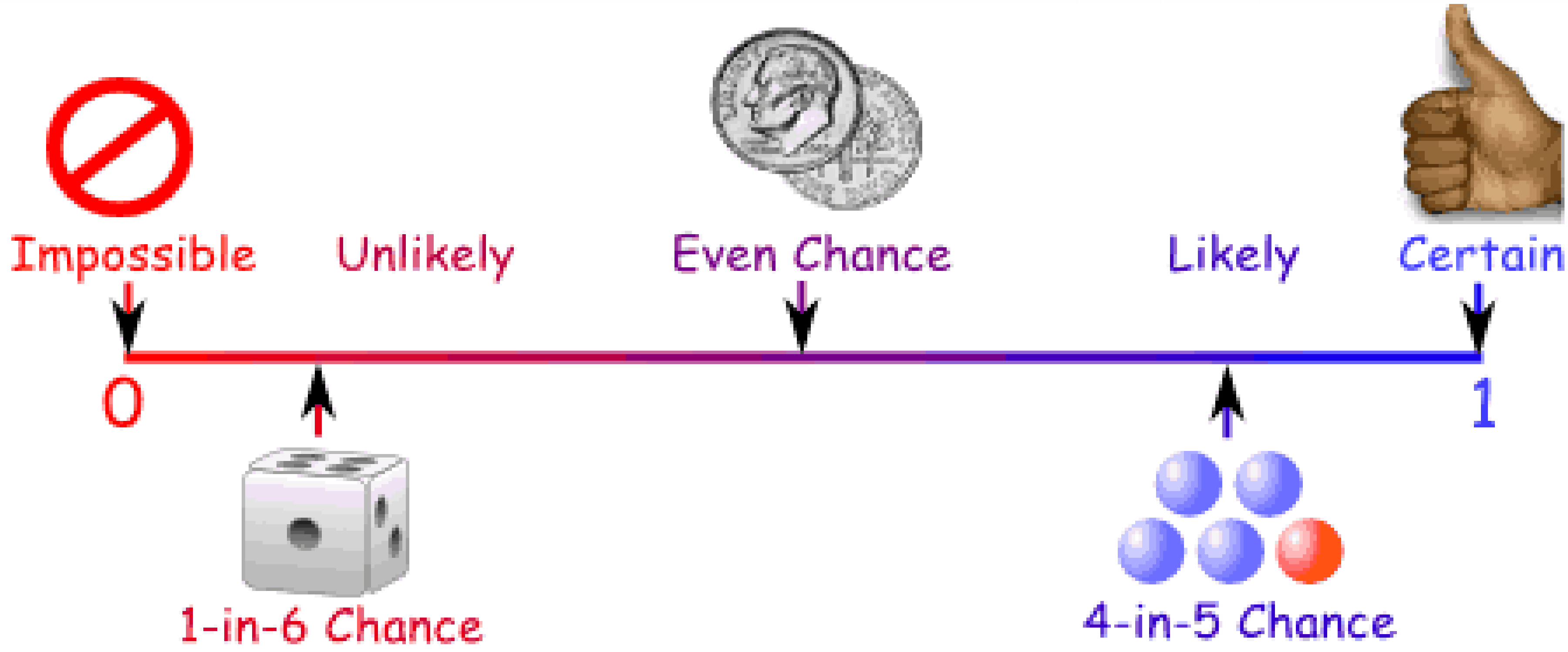
Number of ways it can happen: 4 (there are 4 blues)

Total number of outcomes: 5 (there are 5 marbles in total)

So the probability = 4/5

Probability Line

You can show probability on a Probability Line:



Probability is always between 0 and 1

Probability is Just a Guide

Probability does not tell us exactly what will happen, it is just a guide

Example: **toss a coin 100 times, how many Heads will come up?**

Probability says that heads have a ½ chance, so we would expect 50 Heads. But when you actually try it out you might get 48 heads, or 55 heads ... or anything really, but in most cases it will be a number near 50.

Terminology

Some words have special meaning in Probability:

Experiment or Trial: an action where the result is uncertain.

Tossing a coin, throwing dice, seeing what pizza people choose are all examples of experiments.

Sample Space: all the possible outcomes of an experiment

Example: choosing a card from a deck

There are 52 cards in a deck (not including Jokers)

So the Sample Space is all 52 possible cards:

{Ace of Hearts, 2 of Hearts, etc... }

The Sample Space is made up of Sample Points:

Sample Point: just one of the possible outcomes

Example: Deck of Cards

the 5 of Clubs is a sample point

the King of Hearts is a sample point

"King" is not a sample point. As there are 4 Kings that is 4 different sample points.

Event: a single result of an experiment

<u>Example Events:</u>

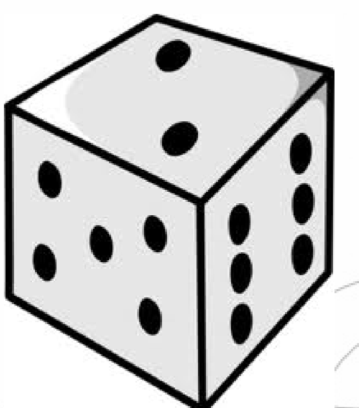
Getting a Tail when tossing a coin is an event

Rolling a "5" is an event.

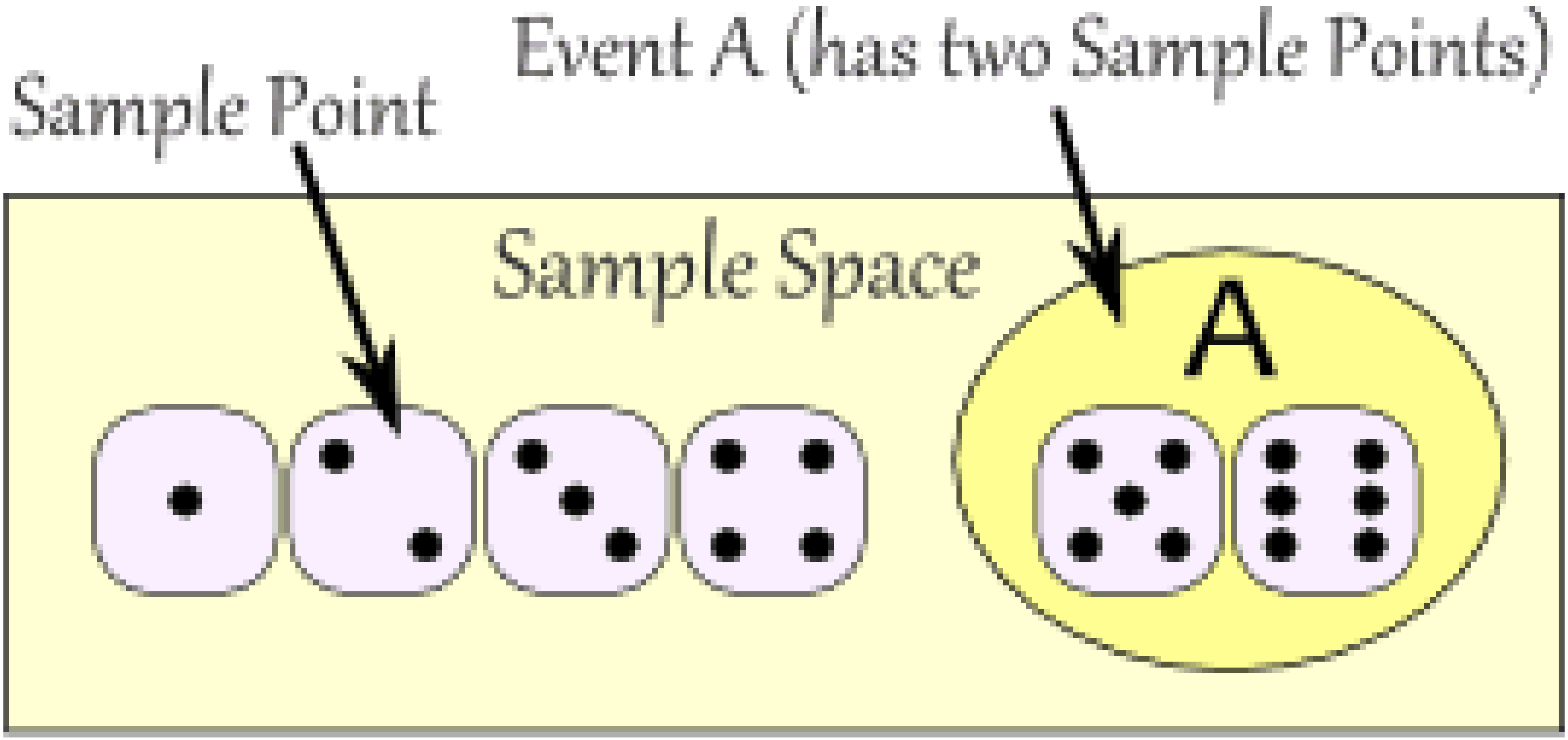
An event can include one or more possible outcomes:

Choosing a "King" from a deck of cards (any of the 4) is an event

Rolling an "even number" (2, 4 or 6) is also an event



The Sample Space is all possible outcomes.
A Sample Point is just one possible outcome.
And an Event can be one **or more** of the possible outcomes.



Mutually Exclusive

Mutually Exclusive means you can't get both events at the same time. It is either one or the other, but **not both**

<u>Examples:</u>
Turning left or right are Mutually Exclusive (you can't do both at the same time)
Heads and Tails are Mutually Exclusive
Kings and Aces are Mutually Exclusive

What isn't Mutually Exclusive
Kings and Hearts are **not** Mutually Exclusive, because you can have a King of Hearts!

Probability: Complement

Complement of an Event: All outcomes that are **NOT** the event.

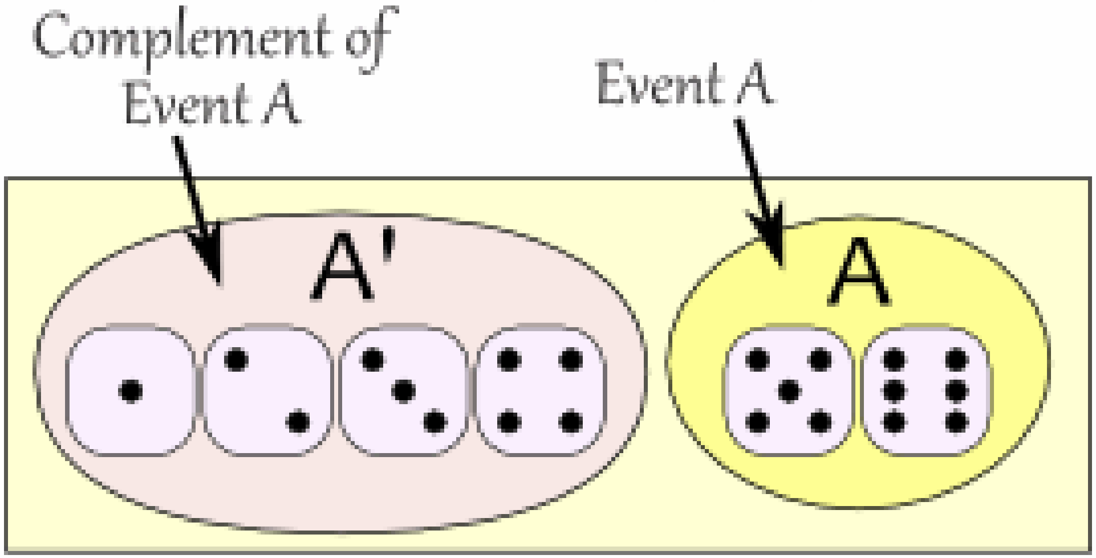
When the event is **Heads**, the complement is **Tails**

When the event is **{Monday, Wednesday}** the complement is **{Tuesday, Thursday, Friday, Saturday, Sunday}**

The probability of an event is shown using "P":
P(A) means "Probability of Event A"

The complement is shown by a little ' mark such as A' (or sometimes A^c):

P(A') means "Probability of the complement of Event A"
The two probabilities always add to 1



$$P(A) + P(A') = 1$$

Independent Events

Events can be "Independent", meaning each event is **not affected** by any other events.

This is an important idea! A coin does not "know" that it came up heads before ... each toss of a coin is a perfect isolated thing.

<u>Example:</u> You toss a coin three times and it comes up "Heads" each time ... what is the chance that the next toss will also be a "Head"? The chance is simply 1/2, or 50%, just like ANY OTHER toss of the coin.

What it did in the past will **not affect** the current toss!

Dependent Events

But some events can be "dependent" ... which means they **can be affected by previous events** ...

Example: Drawing 2 Cards from a Deck

After taking one card from the deck there are **less cards** available, so the probabilities change!

Let's say you are interested in the chances of getting a King.
For the 1st card the chance of drawing a King is 4 out of 52
But for the 2nd card:
If the 1st card was a King, then the 2nd card is **less** likely to be a King, as only 3 of the 51 cards left are Kings.
If the 1st card was **not** a King, then the 2nd card is slightly **more** likely to be a King, as 4 of the 51 cards left are King.

This is because you are **removing cards** from the deck.

Two or More Events

You can calculate the chances of two or more **independent** events by **multiplying** the chances.

So, for Independent Events:

P(A and B) = P(A) × P(B)

Probability of A and B equals the probability of A times the probability of B

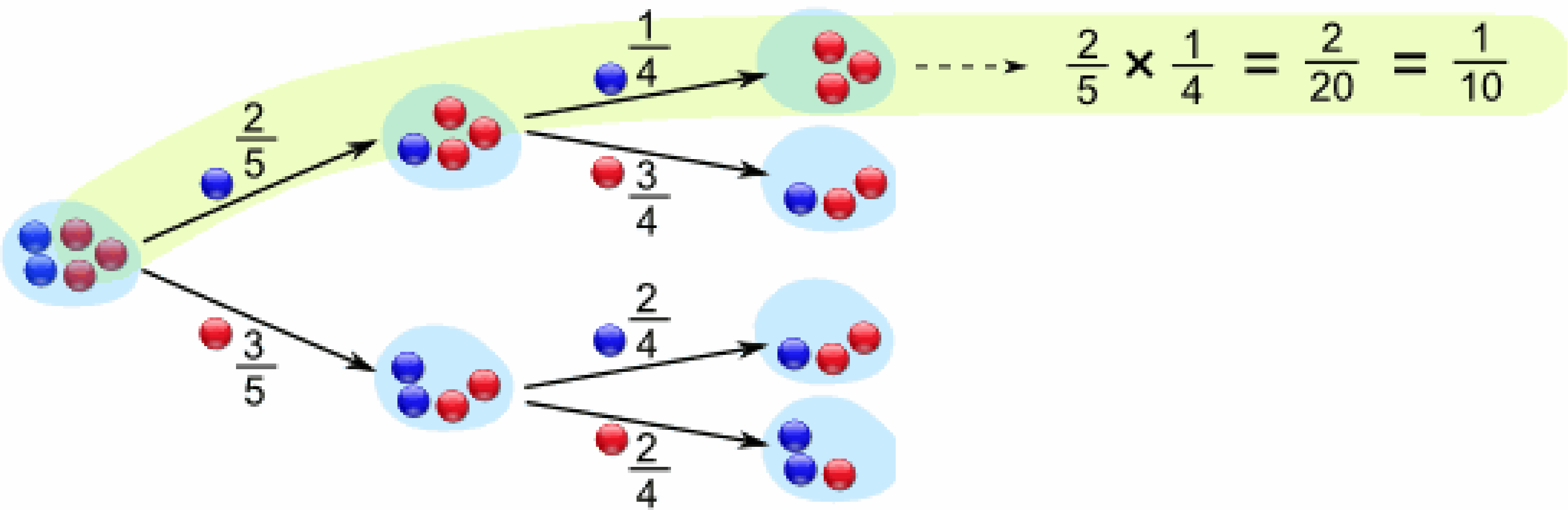
Question 1: What is the probability of 7 heads in a row?

Answer: $\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = 0.0078125$ (less than 1%).

<u>Different</u> for **Dependent** Events!

"What are the chances of drawing 2 blue marbles?"

Answer: it is a **2/5 chance** followed by a **1/4 chance**:



In our marbles example Event A is "get a Blue Marble first" with a probability of 2/5:

P(A) = 2/5

And Event B is "get a Blue Marble second" ... but for that we have 2 choices:

If we got a **Blue Marble first** the chance is now **1/4**

If we got a **Red Marble first** the chance is now **2/4**

So we have to say **which one we want**, and use the symbol "|" to mean "given":

P(B|A) means "Event B **given** Event A“

In other words, event A has already happened, now what is the chance of event B?

P(B|A) is also called the **"Conditional Probability"** of B given A.

Thank You

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